



ESKISEHIR OSMANGAZI UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
2021-2022 SPRING SEMESTER
DIGITAL SYSTEMS LABORATORY

EXPERIMENT 2
COMBINATIONAL LOGIC DESIGN WITH DECODERS AND
MULTIPLEXERS



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GROUP II – D

EXPERIMENT 2 REPORT

OBJECTIVES

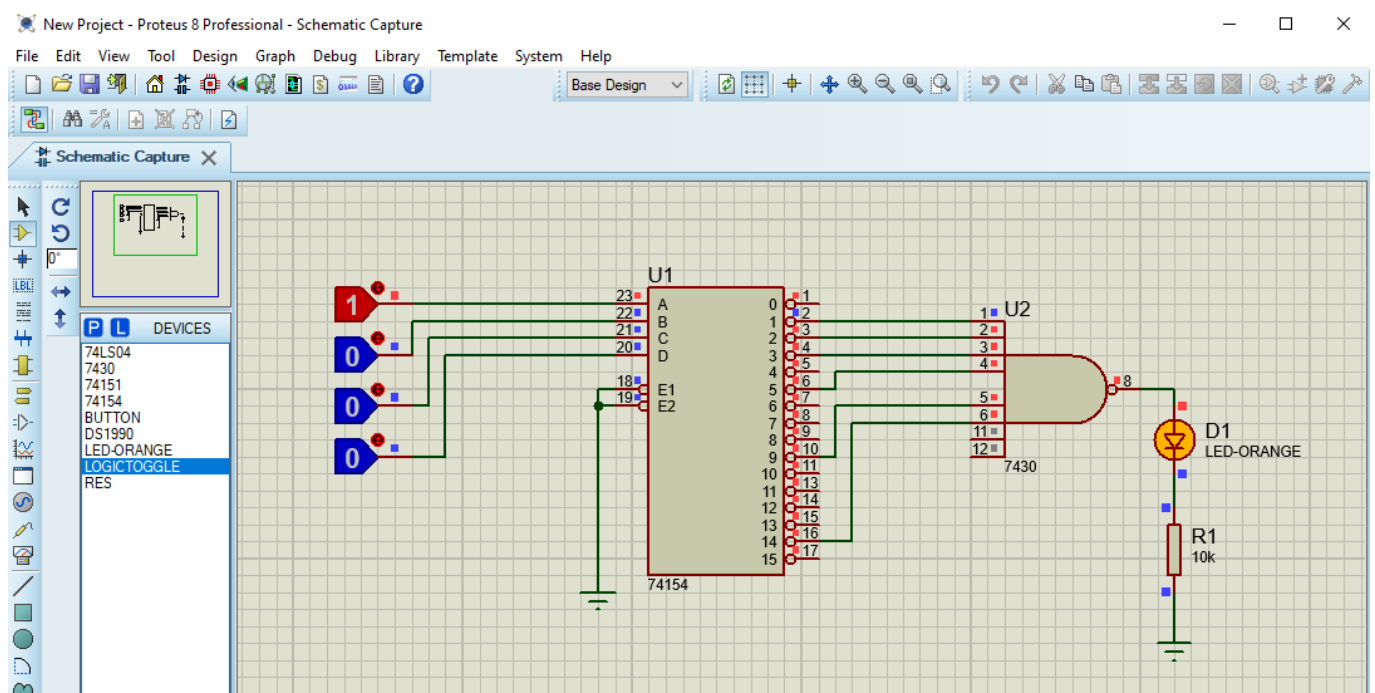
- Designing combinational a circuit an implement it with decoders and multiplexers.

INTRODUCTION

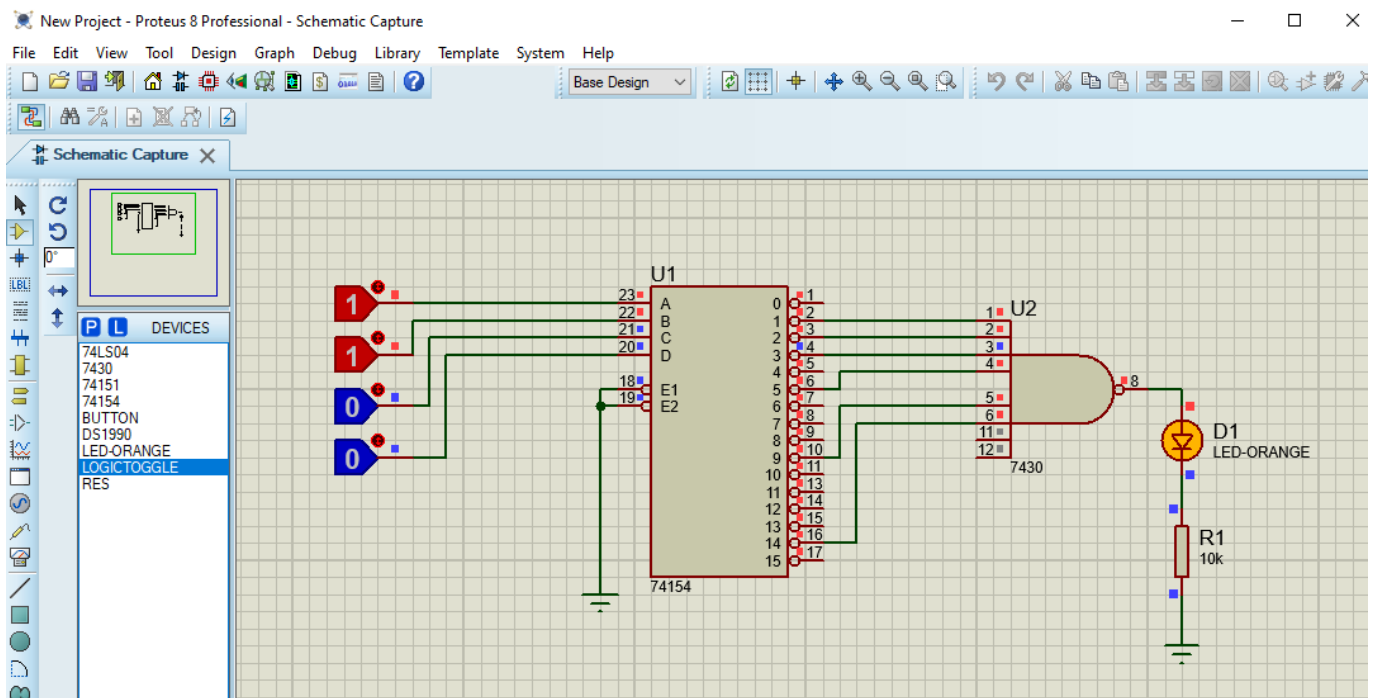
Two circuits were designed in the experiment. In the first phase decoders were used, and multiplexers were used in the second circuit. Circuits were tested with the Fibonacci sequence.

PROCEDURE

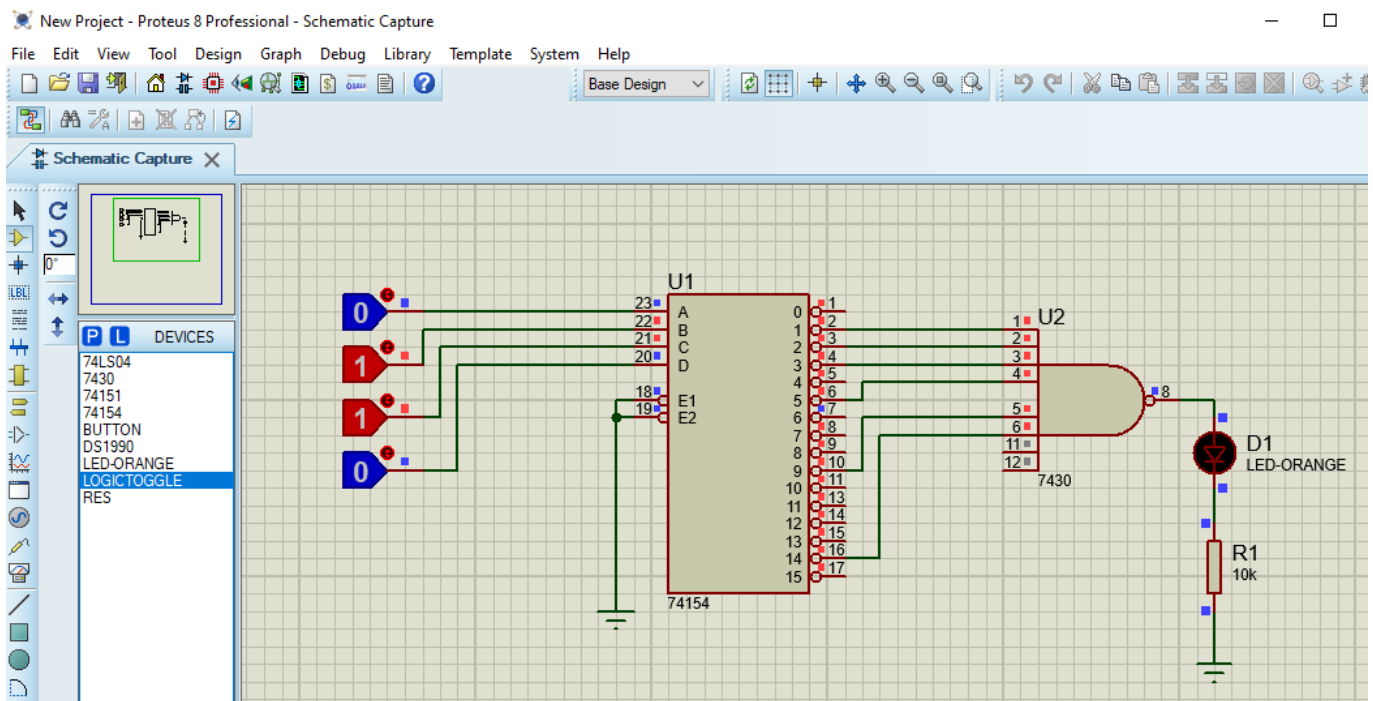
Q1 – Input value 1



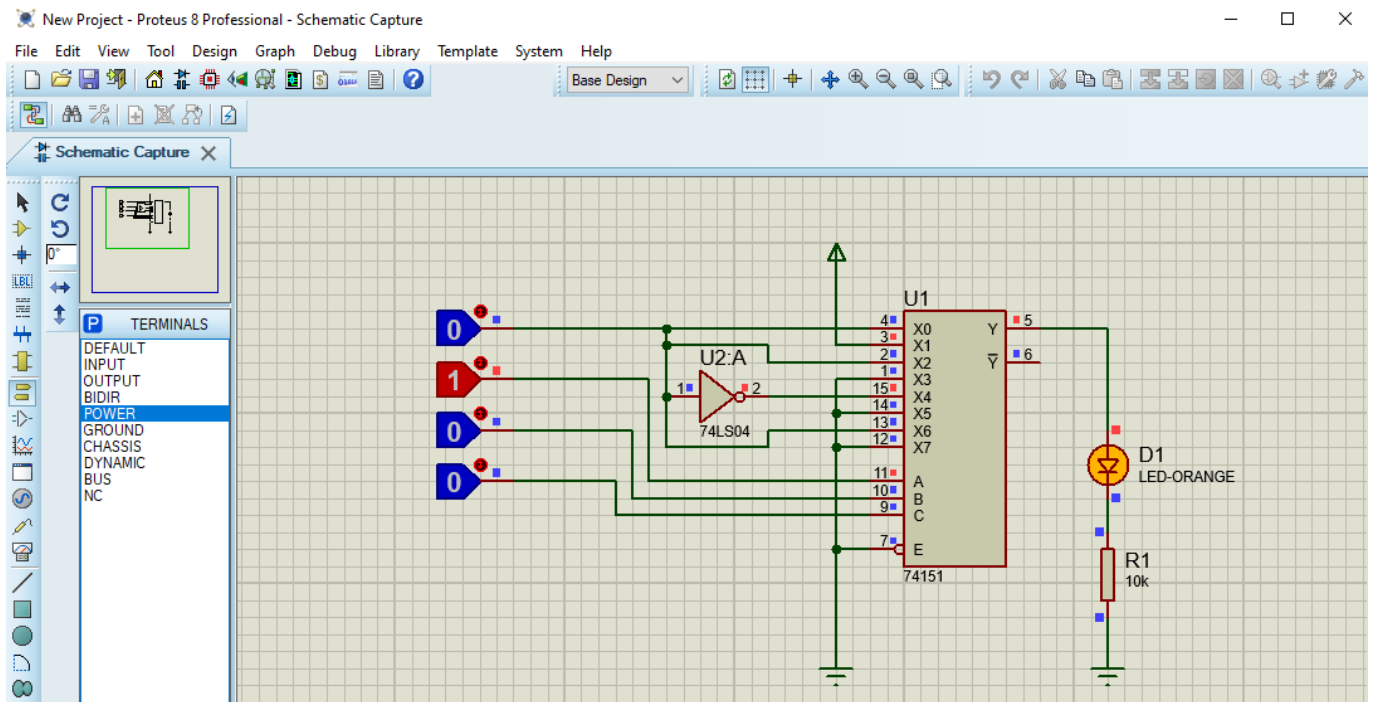
Q1 – Input value 3



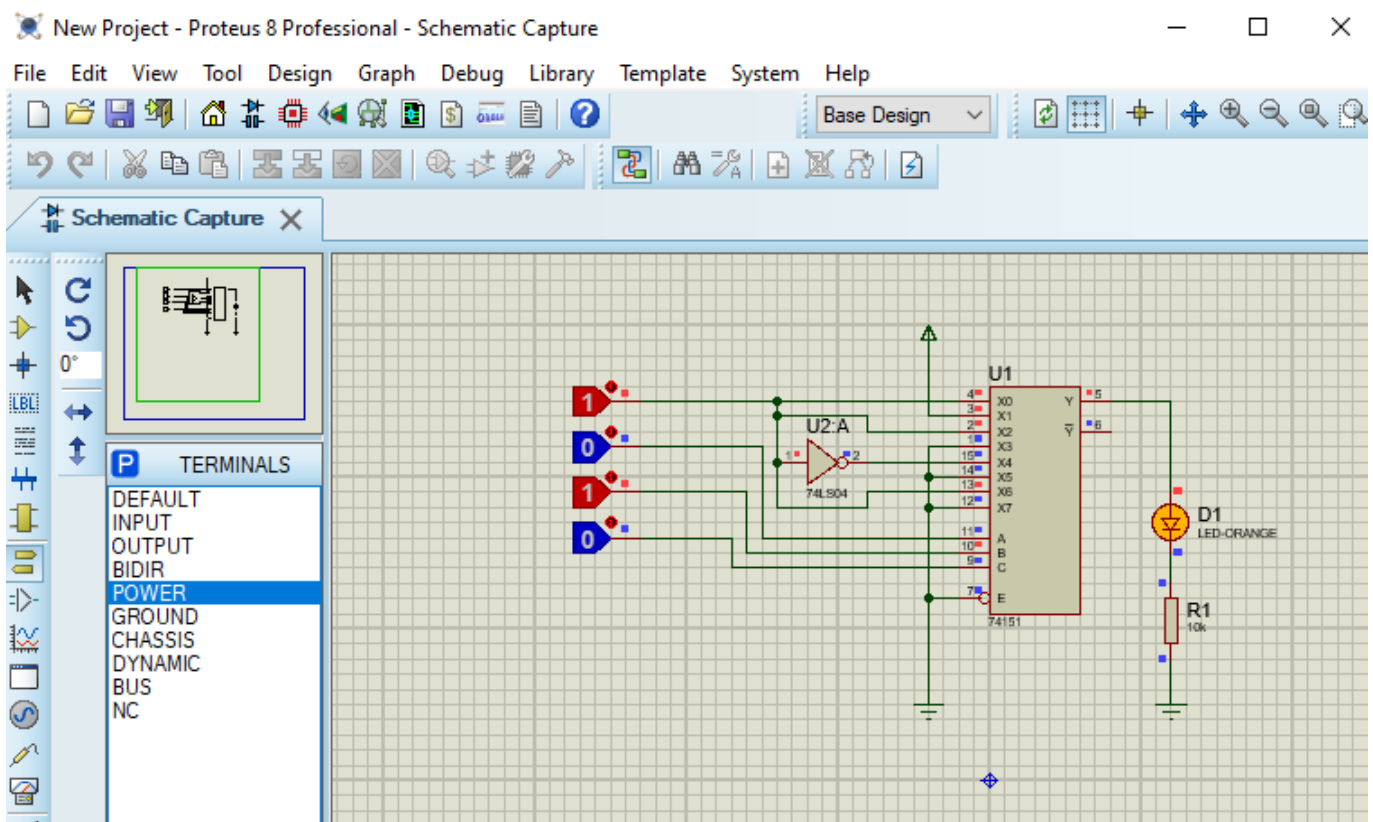
Q1 – Input value 6



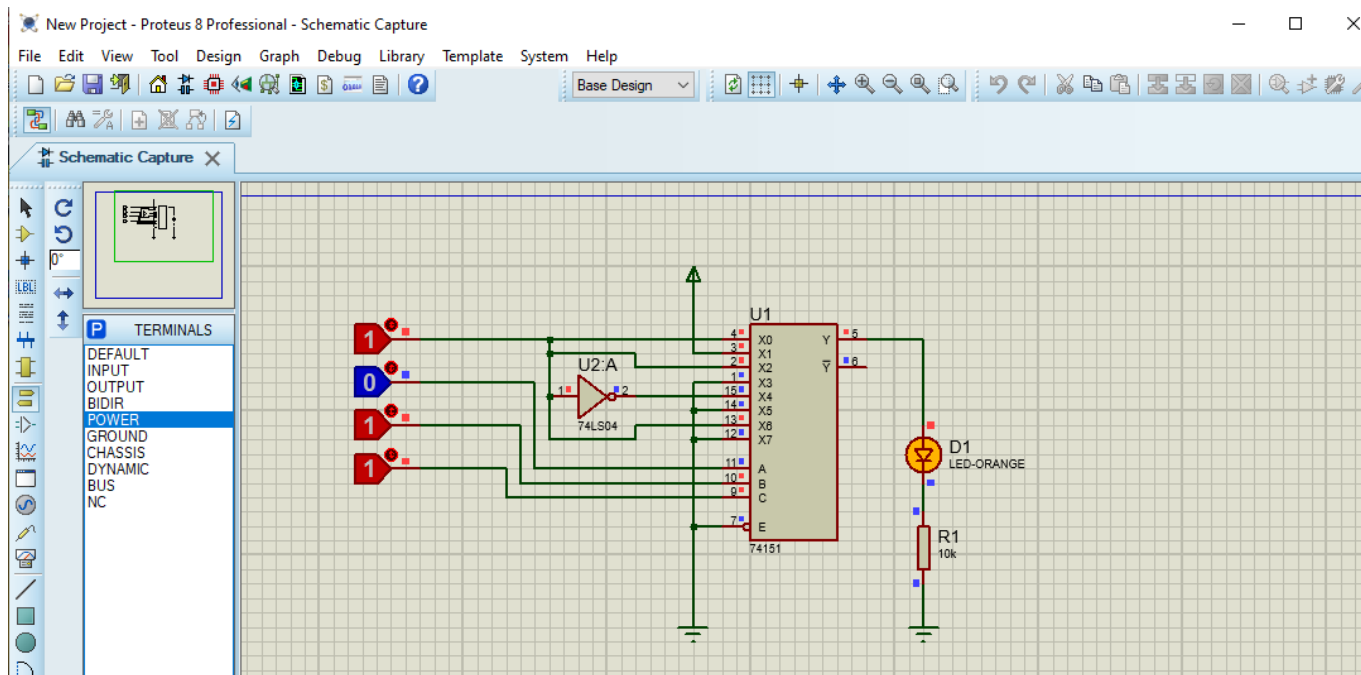
Q2 – Input value 2



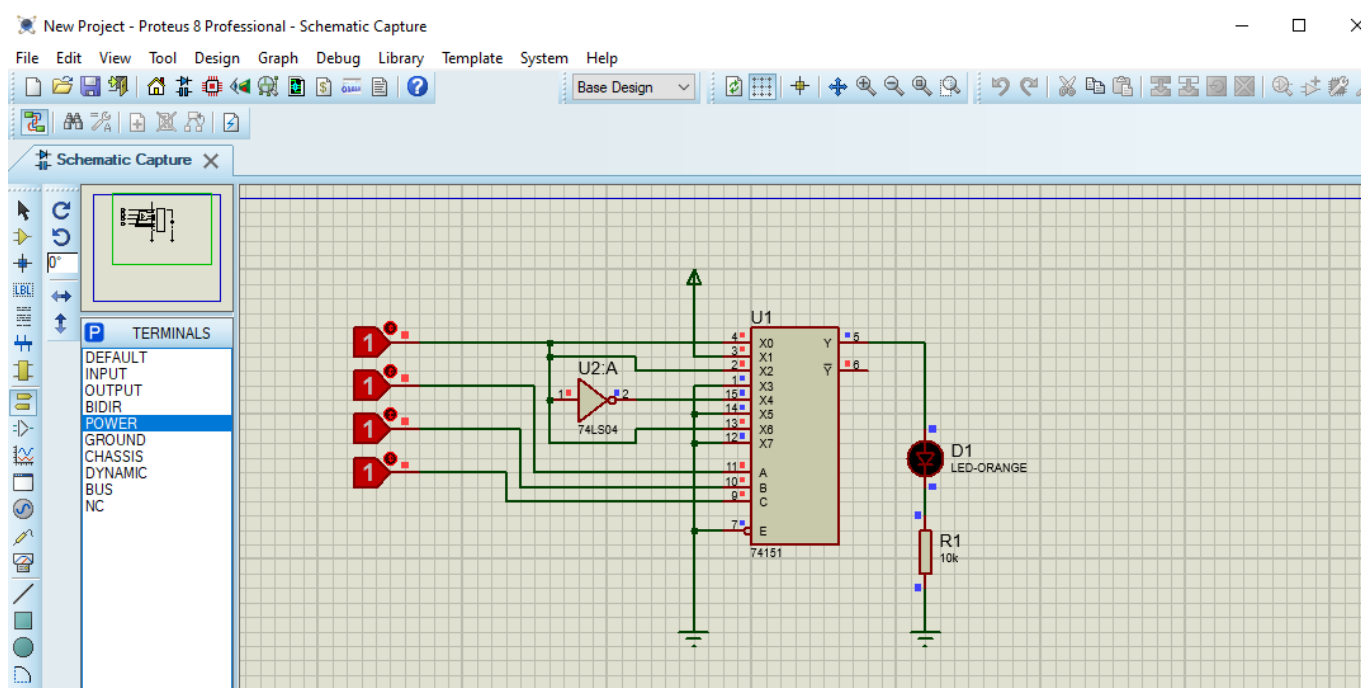
Q2 – Input value 5



Q2 – Input value 13



Q2 – Input value 15



DISCUSSION AND CONCLUSIONS

Since the Fibonacci sequence is a sequence of numbers that is the result of adding each number to the previous one, we entered a bit from the Fibonacci sequence in both circuits in the simulation. If the led is on it means we entered a correct sequence.